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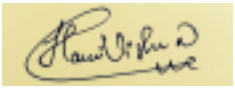
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	SLOT D / 2025	**Date:**	03/10/2026

Code of Conduct

I HARI VISHNU N / 192524319 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem. • Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Annexure A – Analytical Problem Solving

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise). • **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square. • **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]** **(d)** Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]** **(c)** For the hardest and the easiest environment, choose the most suitable agent architecture (simple

reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]** **(d)** Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should

choose. **[5 Marks]**

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

Q1: TWO-ROOM VACUUM WORLD

(a) PEAS Description

To define the agent architecture, the environment is specified using the PEAS framework:

- Performance Measure (□□□□□□□□□):□□
  - ○ + 1 score for every clean square at the end of each time step.
  - ○ − □□ penalty for every movement action (□□□□□□□□□ or □□□□□□ℎ□□).
  - ○ Score formula at time □□:

  - ○ □□□□□□□□□□□□□□□□□□□□□□□□ = (□□□□□□□□□□□□□□□□□□□

○ Combined Percept Vector: ◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆ .

◆◆ = [◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆, ◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆]

(b) Step-by-Step Trace for Agent X (Simple Reflex)

Agent X Decision Logic

- If current square status is ◆◆◆◆◆◆◆◆◆◆◆ → ◆◆◆◆◆◆◆◆◆
- If current square status is ◆◆◆◆◆◆◆◆◆◆◆ → ◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆

Note: Since dirt never reappears, both squares remain Clean at all times after ◆◆ =

3. Total Calculation for Agent X (◆◆ = 1)

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TOTAL	—	—	—	—	—	1	$\sum_{\square\square=1}^{6}$	9

Total Calculation for Agent Y ($\square\square = 1$)

$$\square_{\square\square} = 1 + 0 + 2 + 2 + 2 + 2 = 9$$

(d) Mathematical Derivation of Scores as Functions of

$\square\square$ Agent X Score Breakdown

- Clean square rewards per step:
  - Step 1: $+1$
  - Step 2: $+1$
  - Step 3: $+2$
  - Step 4: $+2$
  - Step 5: $+2$
  - Step 6: $+2$
  - Total Rewards $= 1 + 1 + 2 + 2 + 2 + 2 = 10$
- Movement actions taken: Steps 2 ($\square\square\square\square\square\square h\square\square$), 4 ($\square\square\square\square\square\square\square\square$), 5 ($\square\square\square\square\square\square h\square\square$), 6 ($\square\square\square\square\square\square\square\square$) $\rightarrow 4\square\square\square\square\square\square\square\square\square\square\square\square\square\square\square\square\square\square\square

$$\text{[illegible]}_{\text{[illegible]}}(\text{[illegible]}) > \text{[illegible]}_{\text{[illegible]}}(\text{[illegible]})$$

$$10 - \text{[illegible]} > 10 - 4\text{[illegible]}$$

$$- \text{[illegible]} > - 4\text{[illegible]}$$

- Conclusion: For any positive movement penalty ($\text{[illegible]} > 0$), Agent Y achieves a higher total score than Agent X. If $\text{[illegible]} = 0$, both agents achieve equal scores (10).

(e) Rationality Analysis & Result Interpretation

1. Determination of Rationality
   - Agent Y is Rational, while Agent X is Irrational.

2. Definition of Rationality

An agent is defined as rational if, for every possible percept sequence, it selects an action expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

3. Role of Performance Measure and Percept History
   - The performance measure evaluates the long-term consequence of actions ($+ 1$ per clean square, $- \text{[illegible]}$ penalty for unnecessary movement).
   - Because the environment is partially observable through single-step percepts (an agent in [illegible] cannot perceive square [illegible]'s status directly at $\text{[illegible]} \geq 4$), acting rationally requires information from the percept history.

4. Why Agent X Cannot Match Agent Y (Simple Reflex vs. Model-Based Reflex)

   1. Memoryless Limitation (Simple Reflex):

      At time step $\text{[illegible]} = 4$, Agent X receives the percept [illegible]. Because it maintains no internal state, its mapping is purely condition-action based:

      $$\text{[illegible]}([\text{[illegible]}]) \rightarrow \text{[illegible]}$$

   2. Agent X cannot distinguish between being in square [illegible] for the first time (when [illegible] might be unvisited) versus returning to square [illegible] after both squares have been cleaned. This causes it to oscillate infinitely ([illegible] $\leftrightarrow$ [illegible]), incurring continuous movement penalties ($- \text{[illegible]}$ per step). 3. Internal State Advantage (Model-Based Reflex):

      Agent Y maintains a state memory of past events: [illegible] $= \{\text{[illegible]}, \text{[illegible]}\}$. At $\text{[illegible]} = 4$, even though its immediate percept [illegible] is identical to Agent X's percept, its internal state modifies its decision rule:

   4. This avoids all unnecessary movements, conserving penalties while maintaining maximum clean-square rewards ($+ 2$ per step).

Q2: TASK ENVIRONMENT ANALYSIS AND AGENT SELECTION

(a) Classification of Task Environments

Environment	Property Value	Score	Justification
E1: Autonomous Taxi	Observability Partially Observable	**1**	Sensor range is limited by visual obstacles and weather.
	Number of Agents Multi-Agent	**1**	Interacts with other drivers, pedestrians, and cyclists.
	Determinism Stochastic	**1**	Traffic actions, pedestrian movement, and weather are uncertain.
	Episodic/Sequential Sequential	**1**	Current steering and acceleration choices affect future state sequences.
	Static/Dynamic Dynamic	**1**	Traffic environment continuously changes while the taxi decides its next action.
	Discrete/Continuous Continuous	**1**	Speed, steering angle, and spatial coordinates vary continuously.
E2: Part-Picking Robot	Observability Fully Observable	**0**	Overhead camera captures the entire relevant workspace on the belt.
	Number of Agents Single Agent	**0**	Operates independently on the belt without competing agents.
	Determinism Deterministic	**0**	Belt speed, part locations, and gripper dynamics are fully predictable.
	Episodic/Sequential Episodic	**0**	Picking one part off the belt does not affect how the next part is picked.

	Static/Dynamic Dynamic	**1**	The conveyor belt continues moving while the agent computes its pick action.
	Discrete/Continuous Continuous	**1**	Camera images, joint angles, and velocity trajectories are continuous values.
E3: Chess with Clock	Observability Fully Observable	**0**	Both players have complete visibility of all board positions.

	Number of Agents Multi-Agent	**1**	Involves two competing players maximizing opposite outcomes.
	Determinism Deterministic	**0**	Every move leads to an exact, predictable board state without randomness.
	Episodic/Sequential Sequential	**1**	Early piece openings significantly constrain and dictate mid/late game options.

Static/Dynamic Semi-Dynamic **0.5** Board configuration remains fixed during a turn, but the clock timer runs continuously.

Discrete/Continuous Discrete **0** Finite set of grid squares, clock values, and legal moves per turn.

(b) Difficulty Score Computation and Ranking

Difficulty Score Calculation Table

Environment	**Observability**	**Agents**	**Determinism**	**Episodic/ Sequential**	**Static/ Dynamic**	**Discrete/ Continuous**	**Total Score Calculation**	**Total Difficulty Score**

E1: Autonomous Taxi	1	1	1	1	1	1	1+1+1+1+1+ 1	**6.0**

E2:
Part-Picking Robot

E3:
Chess with Clock
0 0 0 0 1 1 0+0+0+0+1+ **2.0**
1

2.5
5+0

0 1 0 1 0.5 0 0+1+0+1+0.

Environment Ranking

E1 (Autonomous Taxi) [6.0]>E3 (Chess with Clock) [2.5]>E2 (Part-Picking Robot) [2.0]

- 1st (Hardest): E1 (Autonomous Taxi) — Score: 6.0 / 6.0
- 2nd: E3 (Chess-Playing Agent with Clock) — Score: 2.5 / 6.0
- 3rd (Easiest): E2 (Part-Picking Robot) — Score: 2.0 / 6.0

(c) Suitable Agent Architecture Selection

1. Hardest Environment: E1 (Autonomous Taxi)

- Selected Architecture: Utility-Based Agent (or Utility-Based Learning Agent).
- Justification:
  - Stochastic & Multi-agent: In city traffic, there are multiple conflicting goals (e.g., reaching the destination quickly vs. maximizing passenger safety vs. optimizing fuel efficiency vs. maintaining ride comfort).
  - Sequential & Continuous: Simple goal-based agents only evaluate binary success (destination reached: yes/no). A utility-based agent maps state trajectories to real-valued utility trade-offs, enabling optimal decision-making under high uncertainty (stochastic) and dynamic multi-agent conditions.

2. Easiest Environment: E2 (Part-Picking Robot)

- Selected Architecture: Model-Based Reflex Agent (or Simple Reflex Agent with Motion Model).
- Justification:

○ Fully Observable & Deterministic: The robot does not need complex long-term goal planning or utility optimization because each picking task is episodic and independent. ○ Dynamic & Continuous: Because the environment is dynamic (conveyor belt moves continuously), the agent requires a basic state-tracking model to predict the moving part's current coordinate [x(t),y(t)] while calculating continuous joint kinematics. A model-based reflex architecture provides real-time responsiveness without the compute overhead of high-level goal search.

(d) Interpretation of Results

The complexity of an agent program is directly driven by its environment properties rather than its domain name:

1. Observability Drives Memory Complexity: Partially observable environments (E1) require an internal state representation (memory/beliefs) to track unobserved aspects of the world, whereas fully observable environments (E2,E3) allow agents to act purely on immediate sensor percepts.
2. Episodic vs. Sequential Drives Planning Complexity: Sequential environments (E1,E3) require search algorithms, lookahead trees, or dynamic policy evaluations because current actions impact future states. Episodic environments (E2) allow immediate single-step action mapping.
3. Determinism & Multi-Agent Drives Reasoning Complexity: Stochastic and multi-agent dynamics (E1) force the agent to model adversary behavior and handle probabilities, whereas deterministic environments (E3,E2) eliminate the need to plan for uncertain side-effects.

Q3: Turing Test Data Across the History of AI

(a) Correct-Identification Rate and Fooling Rate Computations

Definitions and Formulas

- Total Interrogations (N): 50
- Correct Identifications (C): Number of times the judge correctly identified the machine.
- Correct-Identification Rate (Rcorrect): $R_{correct} = (\frac{C}{N}) \times 100\%$
- Fooling Rate (Rfooling): Percentage of times the machine fooled the judge into believing it was human (N−C). $R_{fooling} = 100\% - R_{correct} = (\frac{N-C}{N}) \times 100\%$

Step-by-Step Calculation Table

System	Era / Technique	Interrogations (N)	Correct Identifications (C)	Correct-Identification Rate Formula	Correct-Identification Rate (Rcorrect)	Fooling Rate Formula	Fooling Rate (Rfooling)

ELIZA	1960s, keyword pattern matching	50	47	(5047)×100 %	94.00%	100%−94.00 %	6.00%

Chatbot B	statistical NLP			28.00%	42.00%
System C	2020s, large neural language model	50 29 (5029)×100 %	58.00%	100%−58.00%	
2010s, scripted rules +	50 36 (5036)×100 %	72.00%	100%−72.00%		

(b) Evaluation Against Turing's 70% Criterion and Chance Level

Method and Threshold Criteria

1. Turing's 70% Criterion: Turing predicted that an average interrogator would have no more than a 70% chance of making the right identification (Rcorrect≤70%). Equivalently, the machine must achieve a fooling rate of at least 30% (Rfooling≥30%).
2. Distance from 50% Chance Level: Calculated as (Rcorrect−50.00%) in percentage points.

Comparative Evaluation Table

System	Correct-Identifi cation Rate (Rcorrect)	Turing Threshold Criterion (Rcorrect≤70%)	Satisfies Turing's 70% Criterion?	Distance from 50% Chance Level Calculation	Distance from 50% Chance Level (Percentage Points)
ELIZA	94.00%	94.00%>70.00%	NO	94.00%−50.00 %	+44.00%
Chatbot B	72.00%	72.00%>70.00%	NO	72.00%−50.00 %	+22.00%

System C		58.00%−50.00 %			
58.00%	58.00%≤70.00%	YES	+8.00%		

(c) Improvement Calculations Across Generations

1. Improvement from ELIZA to System C

Metric	Calculation / Formula	Result
Percentage Points Improvement	Rfooling, System C−Rfooling, ELIZA=42.00%−6.00%	+36.00 percentage points
Ratio Improvement (X-fold)	Rfooling, ELIZARfooling, System C=6.00%42.00 %	7.00× (7-fold increase)

2. Improvement from Chatbot B to System C

Metric	Calculation / Formula	Result
Percentage Points Improvement	Rfooling, System C−Rfooling, Chatbot B=42.00%−28.00%	+14.00 percentage points

ELIZA to System C (Long-term Evolution: 1960s to 2020s):

- Mathematical Gain: The jump from 6.00% to 42.00% represents an absolute increase of 36. 00 and a 7 − (\$600\%\$) relative increase in fooling capability. ● Technical Driver: Represents the paradigm shift from handcrafted, deterministic surface string-matching rules to deep, multi-layer transformer neural architectures trained on massive natural language corpora.

Chatbot B to System C (Mid-term Evolution: 2010s to 2020s):

- Mathematical Gain: The increase from 28.00% to 42.00% yields an absolute gain of 14. 00, corresponding to a 1. 5 − (50%) relative improvement. ● Technical Driver: Illustrates the shift from hybrid systems (combinatorial rule bases paired with localized statistical NLP) to end-to-end generative language models capable of capturing long-range contextual dependencies and fluid conversation structures.

(d) Foundational AI Interpretation: "Acting Humanly" vs. "Thinking / Rationality"

Foundational Concept	Detailed Analytical Explanation	Engineering Relevance

1. The Four Approaches to AI	Artificial Intelligence is categorized into four distinct definitions: 1. *Acting Humanly* (The Turing Test) 2. *Thinking Humanly* (Cognitive Modeling) 3. *Thinking Rationally* (Laws of Thought / Logic) 4. *Acting Rationally* (Rational Agent Approach) The Turing Test explicitly measures Acting Humanly (behavioral imitation). A high fooling rate only demonstrates operational simulation of human surface linguistic patterns, not Thinking or Rationality.	Achieving success on behavioral metrics does not ensure that the system possesses internal cognitive mechanisms, logical reasoning, or valid inference rules.
2. Searle's Chinese Room Argument	John Searle's thought experiment demonstrates that a system can manipulate symbols using syntactic rules (or statistical token probabilities) to produce convincing output without understanding the meaning of those symbols. Application to System C: A Large Language Model (LLM) maps input prompts to high-probability output text via statistical weights ($P(\text{token} \mid \text{tokens}_{1..t-1})$). It processes syntax without semantics.	High fooling rates prove effective statistical behavior matching, not genuine semantic comprehension, intentionality, or rational decision-making.
3. Distinction from Rationality	Rationality requires an agent to take actions that maximize its expected performance measure given its knowledge and percept sequence. Human conversation is full of biases, emotional lapses, and logical errors. Therefore, successfully imitating human speech requires simulating human fallibility rather than acting rationally.	An agent designed to act humanly may intentionally make irrational or logically flawed statements to trick an interrogator, proving that acting humanly and acting rationally can be diametrically opposed goals.

Q4: RATIONAL ROUTE CHOICE UNDER UNCETAINITY

(a) Utility Computation Matrix Across Outcomes

Utility Formula & Conditions

- Utility Function: $\square\square = 100 - \square\square - 20 \cdot \square\square$

- Deadline: ◆◆◆◆◆◆ = 55◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆
- Late Indicator (◆◆):

  $$L = \begin{cases} 1 & \text{if } T > 55 \text{ (Late Trip)} \\ 0 & \text{if } T \le 55 \text{ (On-Time Trip)} \end{cases}$$

Step-by-Step Utility Table

Route	Outcome	Travel Time (T)	On-time Status (T≤55)	Late Penalty (L)	Utility Calculation (100−T−20L)	Utility Value (U)
R1: Highway	Normal	30◆◆◆◆◆◆◆	On-Time	0	100 − 30 − 20	70
	Disrupted	70◆◆◆◆◆◆◆	Late	1	100 − 70 − 20	10
R2: City Road	Normal	45◆◆◆◆◆◆◆	On-Time	0	100 − 45 − 20	55
	Disrupted	60◆◆◆◆◆◆◆	Late	1	100 − 60 − 20	20

R3: Ring Road On-Tim e 50 − 20 50
Normal 50◆◆◆◆◆◆◆ 0 100 − 50 − 20 50 0 100 −
On-Tim e

Disrupted 50◆◆◆◆◆◆◆

(b) Expected Utility Computation and Decision (◆◆)

◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆

◆,◆◆1 = 0. 30

Expected Utility Formula

◆◆[◆◆(◆◆◆◆)] = (1 − ◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆) · ◆◆(◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆) + ◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆· ◆◆(◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆◆)

Expected Utility Table

Route	Normal Prob. (1−P)	Utility Normal	Disrupted Prob. (P)	Utility Disrupted	Expected Utility Formula	Expected Utility E[U]
R1: Highway	0. 70	70	0. 30	10	(0. 70 × 70	
R2: City Road	0. 90	55	0. 10	20	0(0. 90 × 5 51. 5	
R3: Ring Road	1. 00	50	0. 00	50	(1. 00 × 50)	

Optimal Decision

$$\text{E}[\text{U}(\text{R1})] = 52.0 > \text{E}[\text{U}(\text{R2})] = 51.5 > \text{E}[\text{U}(\text{R3})] = 50.0$$

- Decision: The rational agent must choose R1 (Highway) because it yields the highest expected utility (52. 0).

(c) Updated Expected Utility with Radio Alert (E)

E

$P(\text{Alert} \mid \text{Disrupted}) = 0.80$

Recalculated Expected Utility Table

Route	Normal Prob. (1−P)	Utility Normal	Disrupted Prob. (P)	Utility Disrupted	Recalculated Expected Utility Formula	Updated Expected Utility E[U]
R1: Highway	0. 20	70	0. 80	10	(0. 20 × 70) +	22. 0
R2: City Road	0. 90	55	0. 10	20	(0. 90 × 55) +	51. 5
R3: Ring Road	1. 00	50	0. 00	50	(1. 00 × 50) +	50. 0

Updated Optimal Decision

$$\text{EU}[\text{Action}(\text{R2})] = 51.5 > \text{EU}[\text{Action}(\text{R3})] = 50.0 > \text{EU}[\text{Action}(\text{R1})] = 22.0$$

- New Decision: The rational agent changes its choice to R2 (City Road) with an expected utility of 51.5.

(d) Actual Outcome Analysis and Rationality vs. Omniscience

1. Actual Utility Obtained

If the agent chose R1 (as decided in part b) and the accident disruption occurred:

- Actual travel time: $T = 70$ (Late, $L = 1$).
- Actual utility realized: U .

$$U_{\text{Actual}} = 100 - 70 - 20(1) = 10$$

2. Was the Agent Irrational?

- No, the agent was completely rational. Rationality is judged based on the expected outcome at the time the decision was made, given the available information (E), not on the $P_{\text{Actual}} = 0.30$ actual posterior outcome.

3. Rational Agent vs. Omniscient Agent

Dimension	Rational Agent	Omniscient Agent
Definition	Maximizes expected performance given the current percept sequence and prior knowledge.	Knows the actual outcome of every action in advance and maximizes actual performance.

Information Basis	Operates under uncertainty, dealing with probabilities.	Possesses perfect future information ($$ = 0 or $$ = 1 with certainty).
Decision in Part (b)	Selects R1 ($$[$$] = 52. 0), accepting a 30% risk of disruption.	Would select R2 (achieving $$ = 55) or R3 (achieving $$ = 50), knowing R1 will fail ($$ = 10).
Feasibility	Feasible and achievable in real-world AI engineering.	Impossible in non-deterministic, stochastic real-world domains.

(e) Interpretation of Results

Concept	Analytical Explanation
Dependence on Performance Measure	The utility function $$ = 100 − $$ − 20$$ explicitly balances travel duration against a deadline penalty. Changing the weight of $$ or the deadline $$ alters expected utilities and changes the ◆◆◆◆◆◆ optimal route selection.
Dependence on Percept Sequence	Rational action selection is dynamic and depends on the history of sensor inputs up to the current time step. A rational agent updates its internal belief state upon receiving new percepts.

Why Part (c) Changes the Decision

The physical infrastructure of R1 remains unchanged, but the percept sequence received by the agent updated its belief state regarding the disruption probability from 0. 30 to 0. 80. Because expected utility depends on probability beliefs, updating the belief distribution shifts the expected score of R1 from 52. 0

	down to 22. 0, forcing a rational agent to re-adapt its action strategy.

Rubrics for Calculation

Criteria	Weigh ta ge	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understan ding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculatio n	**20% 15%**	Clearly interprets results with units and engineering relevance	errors	minor omissions	No valid calculations
Interpretat ion of Results	All calculations accurate; correct final answer	Minor calculation	Basic interpretation with	Frequent errors; incorrect final answer Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understa nding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculati on (4)	Interpre ta tion of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director

Signature: _____ Signature: _____ Signature: _____ Date: _____ Date: _____ Date: _____